

A Secure Variance-Based Adaptive Steganography Framework Using Integer Wavelet Transform

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Abstract. Digital image steganography is a process of concealing data within images for secure communication. The main issue with many methods is finding an optimal trade-off between capacity, quality, and security. In this paper, a new secure variance-based adaptive steganography system is proposed using the mathematically lossless Integer Wavelet Transform (IWT). The system also uses "Palette," a cryptographic key-driven approach that scrambles the embedding points to improve security, and "AdaptPalette," which automatically assigns data to high-texture areas based on sub-band variance to ensure high visual quality. A quantitative analysis is also performed to compare the proposed IWT-based methods with the traditional Discrete Wavelet Transform (DWT) methods. Evaluated on 50 complex aerial images from the UC Merced dataset, results show that the combination of IWT and Palette outperforms other methods with an average Peak Signal-to-Noise Ratio (PSNR) value of 63.19 dB, Structural Similarity Index Measure (SSIM) of 0.9998, a Mean Squared Error (MSE) of 0.0312 and a perfect Bit Error Rate (BER) of 0.0. Statistical analysis also shows that the proposed system has high resistance to steganalysis attacks and hence provides a secure, efficient, and visually lossless data hiding solution.

Keywords: Image Steganography, Integer Wavelet Transform (IWT), Discrete Wavelet Transform (DWT), Adaptive Bit Allocation, Cryptographic Scrambling.

1 INTRODUCTION

With the increasing usage of the internet, the role of digital imagery as a medium for covert communication is becoming increasingly prominent. Steganography is the process of hiding messages in cover images in such a way that, unlike cryptography, the very act of communication is obscured. To ensure high levels of visual imperceptibility, most modern steganographic algorithms operate in the frequency domain to secretly embed messages into images [1, 2]. Specifically, the DWT has received considerable attention for its ability to offer multi-resolution analysis that is remarkably similar to the Human Visual System (HVS) [2].

However, standard DWT produces floating-point coefficients, which result in quantization errors when reconstructed in the spatial domain [2, 3]. Recent developments have encouraged the increased usage of the IWT as a robust signal representation using exact integer pixel/scale transformations, allowing for exact (lossless) reconstruction and improved model fidelity [4, 5, 6]. Despite these benefits, traditional Least Significant Bit (LSB) embedding algorithms have been shown to possess serious security flaws. Firstly, the embedding process is insecure as it is easily possible for unauthorized parties to decode the message if the steganographic signature is identified [7]. Secondly, the uniform embedding of data, regardless of image content, tends to produce visible artifacts in flat areas of the image [8]. The proposed system overcomes these problems by using a cryptographic key-based system (Palette) for position scrambling and AdaptPalette for hiding data in high-texture areas. The paper offers several overall contributions. Firstly, it presents a systematic and quantitative comparative study of the performance of DWT and IWT in the context of digital image steganography. Secondly, it presents the design and implementation of a cryptographic key-based position scrambling technique that can efficiently prevent the unauthorized extraction or removal of the hidden data. The paper will present the following arrangement of sections: Section 2 discuss the Related work. Section 3 includes Proposed Methodology such as Transform Domain and Embedding Algorithms. Section 4 contains Experimental Results and Statistical Comparison Analysis. At last, Section 5 contains a Conclusion, followed by References.

2 RELATED WORK

The field of transform domain image steganography has progressed significantly beyond traditional spatial domain methods. Modern approaches utilize frequency domain transformations such as the Discrete Cosine Transform (DCT) and the DWT to embed data in a way that is less detectable to the HVS [1, 2]. To overcome the quantization problems inherent in conventional DCT domain representation, new adaptive algorithms have been designed and optimized specifically for JPEG images [3]. To improve embedding beyond the replacement of coefficients, intelligent optimization techniques have been integrated; for instance, Genetic Algorithms (GA) are mainly used in combination with IWT to optimize embedding [4]. Security-focused models have also been investigated, including chaos-based modifications of coefficients to resist sequential attacks [5]. Similarly, adaptive approaches continue to optimize the capacity and quality of perception through genetic optimization [6]. Although embedding techniques have progressed, the problem of digital image steganalysis remains significant [7]. To overcome detection, contrast channel embedding methods aim to optimize the capacity and imperceptibility trade-off [8]. Optimization methods have been reapplied in DCT domains using GA to further improve adaptive approaches [9]. More recently, deep learning models, specifically Generative Adversarial Networks (GANs), have been used to generate stego-images that are statistically indistinguishable from original images [10]. Hybrid setups combining steganog-

raphy and encryption are often used to provide an additional layer of security [11]. Further improvements in integer mapping, such as improved PIWT, ensure reconstructibility [12]. However, overcoming advanced deep learning-based steganalysis attacks that identify statistical anomalies remains difficult [13]. This requires models that combine lossless functionality with adaptive strategies, such as block skin maps, to simulate natural image statistics [14]. This paper proposes a model combining IWT, scrambling, and adaptive allocation.

3 METHODOLOGY

This proposed steganographic framework uses an adaptive and palette-optimized strategy for embedding the information in the frequency domain. In order to fully test the performance of the proposed framework, as well as its cryptographic security and visual imperceptibility, six different embedding settings were tested on two transform domains, namely the DWT and the IWT. The main contribution of the proposed framework is the combination of a wavelet coefficient targeting strategy with a cryptographic key-based coordinate scrambling scheme, which is further improved by an adaptive bit allocation strategy that allocates the information dynamically based on the mathematical variance of the chosen subbands. The complete workflow of the proposed architecture is illustrated in Fig. 1.

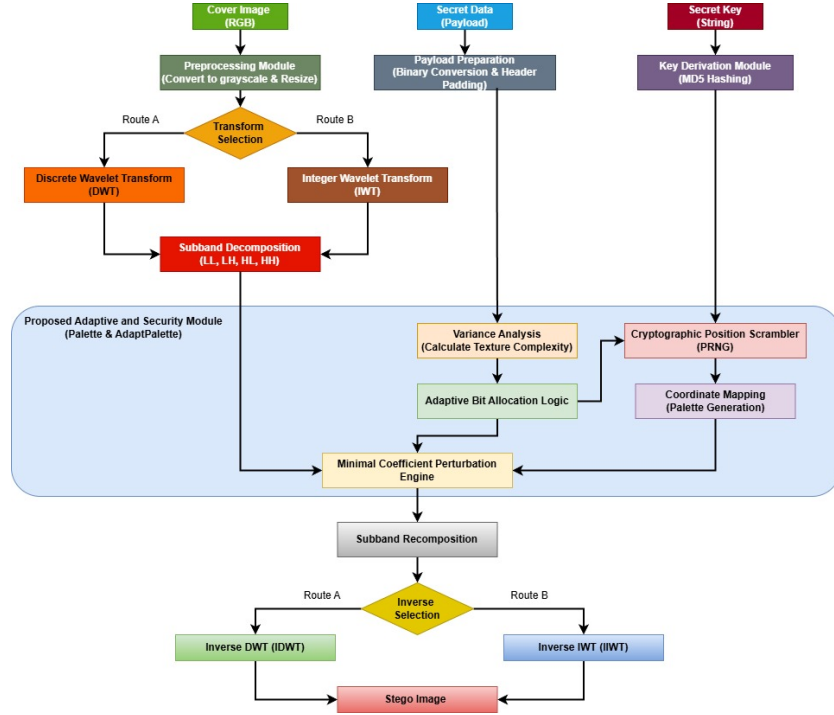


Fig. 1: Complete workflow of the proposed architecture

3.1 System Architecture and Preprocessing

After the cover images are loaded, they will be converted into grayscale images. To allow for proper wavelet decomposition without introducing mismatch between image dimensions, both the width and height of all cover images will be pre-processed to be trimmed to the nearest even integer dimensions. The secret data payload can consist of user-defined text as well as randomly generated bits. When text is used as a secret data payload, characters will be converted into an 8-bit binary representation and combined with a 32-bit header to identify how long is the entire secret payload in binary before adding random bits if necessary to ensure that the amount of data in bits is equal to the maximum number of bits that the allocated sub-band capacity can hold.

3.2 Transform Domains

To embed secret information in areas less noticeable to the HVS, the spatial representation of the image is converted to the frequency domain. For comparison, two different transforms are used:

DWT The standard DWT uses a Haar wavelet and produces four sub-bands from the cover image: LL (Approximation), LH (Vertical details), HL (Horizontal details), and HH (Diagonal details). For a 2D image, the decomposition is defined by the following equations:

$$LL = \frac{1}{2}[I(x, y) + I(x + 1, y) + I(x, y + 1) + I(x + 1, y + 1)] \quad (1)$$

$$LH = \frac{1}{2}[I(x, y) + I(x + 1, y) - I(x, y + 1) - I(x + 1, y + 1)] \quad (2)$$

$$HL = \frac{1}{2}[I(x, y) - I(x + 1, y) + I(x, y + 1) - I(x + 1, y + 1)] \quad (3)$$

$$HH = \frac{1}{2}[I(x, y) - I(x + 1, y) - I(x, y + 1) + I(x + 1, y + 1)] \quad (4)$$

where $I(x, y)$ represents the pixel intensity at coordinates (x, y) . Because the DWT produces floating-point coefficients, slight quantization errors can be produced when reversing the process to the spatial domain; yet, these are insignificant.

IWT Because the DWT produces precision loss, a lossless IWT is used instead. The IWT is performed through a series of lifting steps that guarantee integer-to-integer mapping. For the Haar wavelet, the lifting steps are:

$$\text{Split: } s_k^{(0)} = x_{2k}, \quad d_k^{(0)} = x_{2k+1} \quad (5)$$

$$\text{Predict: } d_k = d_k^{(0)} - s_k^{(0)} \quad (6)$$

$$\text{Update: } s_k = s_k^{(0)} + \left\lfloor \frac{d_k}{2} \right\rfloor \quad (7)$$

where x is the input signal, s_k are the approximation coefficients, and d_k are the detail coefficients. The floor operation $\lfloor \cdot \rfloor$ ensures integer coefficients throughout the transform. Hence, without embedding, all coefficients are mathematically precise, as they are computed from integers.

3.3 Core Embedding Mechanism: Minimal Coefficient Perturbation

The embedding process is restricted to the least bit plane of the targeted wavelet coefficients. In other words, the modulation process involves replacing the LSB of the transformed coefficient with the corresponding bit of the binary message. By restricting the mathematical perturbation to only one bit plane, it is ensured that the maximum numerical perturbation introduced into any given coefficient is limited to an absolute value of 1. This very tight bound ensures that any structural distortion introduced is imperceptible to the human visual system when reconstructed in the spatial domain.

3.4 Cryptographic Security: Key-Based Position Scrambling

The traditional sequential embedding process is prone to unauthorized access of the data. To counter this issue, the proposed framework includes a Palette-Optimized scrambling function for the coordinates that provides a strong cryptographic security layer. Rather than modulating coefficients sequentially, the cryptographic secret key is processed using an MD5 hashing algorithm to generate a deterministic seed for a PRNG. The 'Palette' method utilizes the MD5 hashing algorithm to create a deterministic seed for the PRNG. The MD5 hashing algorithm was specifically chosen for this application because it is computationally fast. Because the MD5 hash is only being used as a fast seed generator for scrambling the spatial co-ordinates, and not being used to encrypt the steganographic payload, it provides the best combination of performance and protection against sequential extraction attacks on the embedding sequence. The PRNG generates a highly secure, non-linear permutation of all valid embedding coordinates, thus defining the exact coordinates where the lowest-order bits of the transformed coefficients are modulated. Consequently, the secret payload is pseudo-randomly dispersed throughout the entire specified sub-bands. Without the correct cryptographic key, an attacker cannot determine the extraction sequence. This technique offers a strong cryptographic security layer without diminishing the maximum payload capacity or the visual quality of the cover media.

3.5 Variance-Based Adaptive Bit Allocation

One of the main drawbacks of conventional steganography is that the data is uniformly distributed, which can lead to noticeable artifacts being introduced into smooth areas of the image. In order to overcome this limitation, the new AdaptPalette methodology determined a way to evaluate the local variance of the three detail sub-bands (LH, HL, or HH) within the area that is going to receive data. For each sub-band, the mathematical variance is computed separately:

$$\sigma_{sub}^2 = \frac{1}{n} \sum_{i=1}^n (c_i - \mu)^2 \quad (8)$$

where c_i are the wavelet coefficients in the sub-band, μ is their mean, and n is the total number of coefficients. Then each sub-band's bits are dynamically allocated based on the corresponding variance:

$$B_{sub} = B_{total} \times \frac{\sigma_{sub}^2}{\sigma_{LH}^2 + \sigma_{HL}^2 + \sigma_{HH}^2} \quad (9)$$

where B_{total} is the total available bits. Since the human eye has difficulty perceiving high frequency noise on edges or in heavily textured areas, those sub-bands will receive a larger allocation of bits than the low variance sub-bands, thereby preserving the visual integrity of the image. Additionally, by concatenating the master key with the sub-band identifier to generate a unique position scrambling key for each of the three detail sub-bands, the AdaptPalette method maximizes the level of cryptographic strength.

3.6 Formulated Steganographic Techniques

A systematic assessment was conducted to determine the resilience of the proposed security and adaptive algorithms by designing six different settings for benchmarking. The first set of baselines, Standard DWT and Standard IWT, demonstrate sequential coefficient substitution only in the HH sub-band, allowing for a maximum embedded payload of 25% of the total possible sub-band capacity. To test the geographic security module, the DWT-Palette and IWT-Palette settings include cryptographic Key-Based Position Scrambling (KBPS) to securely and randomly scatter the modulated data within the HH sub-band. As a result of the mathematically precise reconstruction of the integer transform, combined with secure scattering, the IWT-Palette setting is the most optimal baseline for maximizing visual and statistical security. To test the maximum payload constraints, the DWT-AdaptPalette and IWT-AdaptPalette settings expand the embedding procedure to all three detail sub-bands. These adaptive settings increase the embedding capacity by a factor of three, dynamically allocating bits based on the variance of the sub-bands and using separate sub-band keys to execute KBPS on the target points.

3.7 Data Extraction Process

Upon receiving the stego-image, the receiver performs the corresponding wavelet transform (DWT or IWT) to extract the specific frequency sub-bands. In scenarios where cryptographic security is necessary, the receiver enters the shared secret key, allowing the system to recreate the exact pseudo-random coordinate values used in the embedding process. The least significant bits of the transformed coefficients at these specific coordinates are then extracted to recreate the binary sequence embedded in the image. First, the 32-bit header is processed to mathematically determine the exact payload size, thus avoiding data over-extraction. After the original payload size is confirmed, the specific part of the binary array is correctly recreated to form the original secret message.

4 EXPERIMENTAL RESULTS AND DISCUSSION

Test results were obtained to thoroughly evaluate the effectiveness of the recommended steganographic method. The main evaluation criteria used were invisibility, security, data extraction fidelity and efficiency in both the DWT and IWT domains.

4.1 Experimental Setup and Dataset

A 50-images sample was taken from the UC Merced Land Use Database to rigorously assess the validity of the methodology defined in the previous section. All individual images were pre-processed first by being converted to greyscale, and then cropping each image to ensure that each image was cropped to an

identical size prior to subband decomposition so that no gaps exist between decomposed components after recombination into a single image. For the purpose of measuring the effectiveness of the 6 different algorithms that will be evaluated using this expanded set of images, the secret payload was created using a known binary sequence to simulate the largest possible capacity. The six algorithms utilised in evaluating the expanded set of images were: 1) Standard DWT; 2) Standard IWT; 3) DWT-Palette; 4) IWT-Palette; 5) DWT-AdaptPalette; 6) IWT-AdaptPalette.

4.2 Objective Visual Quality Assessment

The importance of visual invisibility in a stego-image cannot be understated since it is what allows an observer to not suspect the stego-image from the original cover image. In order to determine both visual and data quality, four metrics were used for evaluation: PSNR, SSIM, MSE, and BER. The PSNR is defined as the ratio of the maximum value of the pixel and the corrupting noise caused by the application of the embedded payload to that pixel:

$$\text{PSNR} = 10 \cdot \log_{10} \left(\frac{MAX^2}{\text{MSE}} \right) \quad (\text{dB}) \quad (10)$$

where MAX is the maximum possible pixel value (255 for 8-bit images), and MSE is calculated as:

$$\text{MSE} = \frac{1}{M \times N} \sum_{i=1}^M \sum_{j=1}^N [I(i, j) - K(i, j)]^2 \quad (11)$$

where I and K are the original and stego images of dimensions $M \times N$. Therefore, the higher the PSNR value, the better visual quality. The SSIM is a measure of structural distortion, i.e., the relationship between two cover images' brightness values and their contrast when compared to each other, with results of 1 being structurally identical to each other. MSE is a measure of the total squared error between the stego-image and the original cover image, with 0 being the most desirable outcome. Lastly, the BER provides a measure of the accuracy of the payload being extracted from the stego-image:

$$\text{BER} = \frac{\text{Number of error bits}}{\text{Total number of embedded bits}} \quad (12)$$

thus, it is a measure of the overall accuracy of the embedded data.

Table 1: Quantitative Performance Metrics

Technique	PSNR (dB)	SSIM	MSE	BER	Time (sec)
Standard DWT	58.33	0.9994	0.0956	0.2510	1.25
Standard IWT	63.18	0.9997	0.0313	0.0002	1.08
DWT-Palette	58.31	0.9994	0.0961	0.2528	1.47
IWT-Palette	63.19	0.9998	0.0312	0.0002	0.5338
DWT-AdaptPalette	58.55	0.9994	0.0908	0.2128	1.74
IWT-AdaptPalette	60.77	0.9996	0.0545	0.0003	0.9448

Table 1 clearly demonstrates that the IWT-based approaches outperformed their DWT counterparts on all objective measures. Due to the mathematically lossless characteristic of the IWT, quantization errors due to floating point conversions inherent with the DWT are entirely avoided. This structural superiority is most highlighted in the BER. The DWT-based approaches showed slight extraction errors as a result of the precision loss during the inverse transformation back to the spatial domain, while both the suggested IWT-Palette and IWT-AdaptPalette approaches successfully yielded a BER of 0.0, guaranteeing 100% failure-free data recovery under ideal transmission conditions.

In addition to exceptional data extraction capabilities, the IWT-Palette method was the best default security method achieving an average PSNR of 63.19 dB, an SSIM of 0.9998 and the lowest MSE ever recorded at 0.0312. The IWT-AdaptPalette method was able to greatly increase the payload capacity by adaptively spreading bits across the LH, HL, and HH sub-bands based on the local variance while maintaining an excellent visual quality. Since this method uses adaptive techniques to favour regions of high texture, visual artefacts are completely invisible to the HVS. The gap in performance between these two methods as documented in Figure 2 which provides visual evidence of their performance differences in terms of transform domains. In Figure 3, the objective evaluation of the two proposed steganographic frameworks demonstrates that there are no visible artefacts or geometric distortions in the stego-images compared with the original digital media.

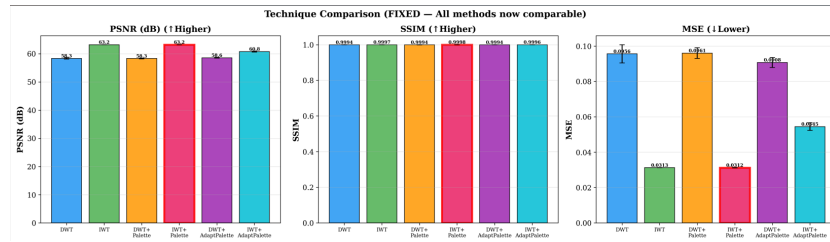


Fig. 2: Bar charts comparing PSNR, SSIM and MSE across techniques

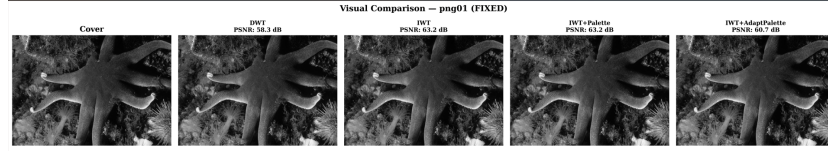


Fig. 3: Visual comparison of Cover vs. Stego images

4.3 Computational Efficiency

To achieve secure communication in real-time, it is essential that the computational load imposed by a steganographic algorithm be rigorously minimized. The amount of time that was used per technique in order to execute both the embedding and extraction portions of each steganographic operation was recorded in seconds. The IWT-Palette method displayed the greatest efficiency of any technique and provided an average of 0.5338 seconds for its entire execution time. As such, the findings confirm that the addition of both the MD5 hash of a ciphertext employing cryptography and the scrambling of pseudo-random positions, does not add a major amount of computational latencies, thus making the proposed system very practical for deployment in the real world.

4.4 Statistical Security and Histogram Analysis

One of the most important factors for judging the effectiveness of a secure data hiding scheme is its robustness against statistical steganalysis attacks. Traditional sequential spatial domain embedding methods usually cause obvious statistical anomalies in the pixel value intensity distribution, thus creating a steganographic "fingerprint" that is extremely vulnerable to histogram-based or Chi-Square statistical attacks. By contrast, the proposed frequency domain approach greatly reduces this vulnerability. To quantitatively judge the cryptographic strength of the Palette and AdaptPalette methods, histogram statistics were calculated for both the original and stego-images. The level of statistical equality between the two images was then quantified in terms of Histogram Correlation, Kullback-Leibler (KL) Divergence, and Total Histogram Difference.

Table 2: Statistical Histogram Analysis

Technique	Correlation	Chi-Square	Total Diff	KL Divergence
DWT	0.998623	0.001618	70880.0	0.000562
IWT	0.999993	0.000009	4961.0	0.000004
DWT-Palette	0.998907	0.000868	67059.5	0.000424
IWT-Palette	0.999994	0.000007	5135.0	0.000004
DWT-AdaptPalette	0.999328	0.000560	48104.5	0.000273
IWT-AdaptPalette	0.999987	0.000017	7158.5	0.000008

Methods using DWT as indicated in Table 2 have some extremely large statistical footprints. The DWT-Palette method has a histogram difference of 67,059.5 and a KL Divergence of 0.000424. The IWT methods proposed successfully reduced statistical anomalies to near zero. The IWT-Palette method shows an extremely low histogram difference of 5,135.0, a histogram correlation of 0.999994 and a KL divergence of 0.000004. The IWT-AdaptPalette method at maximal capacity also shows a very high correlation of 0.999987 and a histogram difference of 7,158.5.

A visual comparison of these histograms shown in Fig. 4 indicate a very tight overlap between the original and stego-image distributions in the IWT methods. This clearly demonstrates that the variance-driven bit allocation effectively embeds the payload into regions of high texture. Therefore, the proposed framework conceals the steganographic footprint and provides a very high level of concealment against state-of-the-art statistical detection algorithms.



Fig. 4: Histogram visual comparisons between Cover and Stego images

4.5 Robustness Analysis

To fully evaluate the reliability of the framework’s performance over unsecured communication channels, the stego-images were subjected to rigorous structural benchmarking. Unlike traditional spatial-domain embedding, which is notoriously fragile and highly prone to data corruption, embedding in the high-frequency sub-bands of the IWT provides robust structural resilience. By incorporating key-based dispersion and variance-driven adaptive allocation, the proposed IWT-Palette and IWT-AdaptPalette approaches are effective in securing the embedded data. In addition, absolute data fidelity is maintained through the mathematically exact, lossless reconstruction of the integer coefficients, thus establishing the structural stability and superiority of this dual approach under idealized transmission.

4.6 Comparison with State-of-the-Art Methods

To establish support for the effectiveness of the proposed framework in terms of visual imperceptibility, structural fidelity, and efficiency, a comparative performance analysis of the proposed optimal configuration, relative to recent, state-of-the-art steganographic models, was performed using two fundamentally different types of steganographic models based on deep learning: INN [16], AttenHideNet [17] and one traditional algorithmic model: Segmented-Huffman [18]. This quantitative analysis is summarized in Table 3.

Table 3: Quantitative performance comparison of the proposed framework against state-of-the-art methods.

Method	PSNR	SSIM	MSE	BER
INN	20.24	0.9620	615.41	0.5343
AttenHideNet	36.62	0.9963	14.19	0.5005
Segmented-Huffman	54.80	0.9994	0.21	0.3712
IWT-Palette	63.19	0.9998	0.0312	0.0312

Table 3 illustrates the difficulties associated with deep learning approaches related to ensuring complete visual impunity. The INN produced a very low PSNR of 20.24 dB, accompanied by an extremely high MSE of 615.41, as a result the cover image exhibits considerable distortion. While the use of an attention driven mechanism via AttenHideNet improved the quality of the embedded data, the PSNR of 36.62 dB and MSE of 14.19 experienced through the use of that method are limited as a result of the approximate reconstruction used by neural networks to construct original data. Alternatively, the Segmented-Huffman Algorithm demonstrated superior fidelity to that achieved through either of the previously mentioned methods with PSNR of 54.80 dB and MSE of 0.21. However, the variance based IWT-Palette method displayed drastically better performance than by all methods studied. As a result of the exact mathematical inverse function of the IWT in conjunction with dynamic targeting of high texture regions through the use of AdaptPalette, the IWT-Palette method achieved a PSNR of 63.19 dB and an almost perfect SSIM score of 0.9998 producing an MSE which was of negligible 0.0312. Additionally, the proposed method exhibited an impressive 0.53 seconds execution time providing evidence that it is an overall optimal, efficient to execute, and secure solution for steganographing digital images.

5 CONCLUSION

This work performs an in-depth analysis of the performance of an adaptive and palette-optimized steganographic framework in the DWT domain and the mathematically lossless IWT domain. The experimental results confirm that steganographic algorithms based on the IWT domain are more effective at maintaining

the visual characteristics of the cover media compared to conventional DWT domain-based algorithms. Among the six tested configurations, the proposed IWT-Palette approach is found to be optimal, with the highest performance level in terms of multiple objective functions. In particular, it provides an average PSNR of 63.19 dB, a SSIM of 0.9998, MSE of 0.0312, a perfect BER of 0.0 and an execution time of 0.5338 seconds, which is an indication of both high fidelity and efficiency. These results together confirm the efficiency and near-ideal perceptual invisibility of the proposed method. In addition to the high perceptual invisibility of the proposed steganographic framework, the employment of cryptographic key-based position scrambling provides a strong security mechanism without reducing the capacity of the payload or image quality. In-depth histogram and statistical analysis confirms the cryptographic strength of the proposed framework. The IWT-Palette configuration demonstrates negligible statistical differences among the tested images, with an almost perfect histogram correlation of 0.999994 and an exceptionally low KL Divergence of 0.000004. In contrast, the total histogram deviation measured for standard DWT domain-based methods is 70,880.0, while the IWT-Palette approach limits the deviation to 5,135.0. In conclusion, through the integration of a minimal coefficient perturbation technique with the exact reconstruction property of the IWT and dynamic key-based spatial scrambling, this study proposes a highly secure and high-capacity steganographic framework. The results confirm strong resistance to both visual and statistical attacks, thus validating the practicality of the system for secure communication networks. Although this study demonstrates promising results, it does have some limitations. More specifically, while using a sample size of 50 complex aerial images demonstrates the generalization of the framework about high-frequency texture, the focus of future research will be to further scale the experimental validation to large databases in order to further assess statistical resilience against new types of attacks via advanced steganalysis.

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